A Rotation Creates the Term $P \oplus Q$

		Pauli String	Qubit 1	Qubit 2	Qubit 3	x Bits 1	x Bits 2	x Bits 3	z Bits 1	z Bits 2	z Bits 3	Row r = Parity of the z Bits
	P	Term	Z	I	I	0	0	0	1	0	0	$r(P) = 1$
$\oplus$	Q	Generator of an X Rotation on Qubit 1	X	I	I	1	0	0	0	0	0	$r(Q) = 0$
=	$P \oplus Q$	New Term	Y	I	I	1	0	0	1	0	0	$r(P \oplus Q) = 1$

Only This x Bit Changed

z Bits Unchanged

If $r(Q) = 0$ Then $r(P \oplus Q) = r(P)$